MIL-A-22789C
1 April 1981
SUPERSEDING
MIL-A-22789B
11 February 1974

MILITARY SPECIFICATION

ANKLETS, WOMEN'S: COTTON

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for one type of cotton anklet.

1.2 Classification The anklets shall be of one type and of the following classes and sizes (see 6.2):

*1.2.1 Classes Class 1 - White
Class 2 - Black

*1.2.2 Sizes 8 1/2, 9, 9 1/2, 10, 10 1/2, 11 and 11 1/2

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-B-17757 - Box, Shipping, Fiberboard (Modular Sizes)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8445

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:59Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1491	- Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

*2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained from the Federal Trade Commission, Washington, DC 20580).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

NATIONAL MOTOR FREIGHT CLASSIFICATION

(Application for copies should be addressed to American Trucking Association, Inc., Attn: Traffic Department, 1616 P Street, NW, Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, ILL 60606).

3. REQUIREMENTS

3.1 Samples

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:59Z
Check the source to verify that this is the current version before use.

3.1.1 Guide samples Sample when furnished is solely for guidance and information to the contractor. Variations from the specification may appear in the sample in which case this specification shall govern (see 6.3).

3.1.2 Standard sample The anklets shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

*3.2 First article When specified, (see 6.2), the contractor shall furnish a sample unit(s) for first article approval (see 4.3).

3.3 Material

3.3.1 Yarn, knitting The yarn used for knitting the anklets shall be cotton, combed, mercerized and shall be size 30/2 (± 3%).

3.3.2 Reinforcing yarn Yarn used for reinforcing the heel and toe shall be filament nylon and of denier suitable for the intended purpose.

3.3.3 Looping yarn The looping yarn used for the toe closure shall be cotton, mercerized, 2 or 3 ply. As an alternate, nylon thread of a suitable size may be used for sewing the toe closure.

3.3.4 Welt yarn The top welt yarn shall be a cotton-covered rubber yarn of a size suitable for the intended purpose.

3.4 Construction The anklets shall be seamless, circular knit, 48 gauge, crew type. The leg and foot shall be a 2 by 2 rib knit, using yarn specified in 3.3.1. The sole, ring toe and heel shall be a plain knit. The toe and heel areas shall be reinforced using yarn specified in 3.3.2. The top welt shall be non-ravelling and shall include one strand of yarn as specified in 3.3.4. The anklets shall be knitted on a 3, 3 1/4 or 3 1/2 inch cylinder, using 167 to 186 needles per size. The toe closure shall be as elastic as the knit portion of the anklet and shall be free of lumps, loops or other defects that affect serviceability. The free ends of the looper or sewing thread shall be a minimum of 1/2 inch and a maximum of 1 inch. The toes, sewed with yarn specified in 3.3.3, shall be seamed inside-out, using seam type SSa-1 and stitches types 503, 504, or 505 of FED-STD-751, with 50 to 55 stitches per inch in the relaxed state.

3.4.1 Finished weights The finished anklets shall conform to the following minimum conditioned weights when tested as specified in 4.4.4.

Sizes	Weight (ounces per dozen)
8 1/2	9
9	10
9 1/2	10 1/2
10	11
10 1/2	11 1/2
11	12
11 1/2	12 1/2

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:59Z
Check the source to verify that this is the current version before use.

3.5 Color Class 1 anklets shall be bleached white and may be supplemented with a fluorescent brightener which fluoresces in the blue region. Class 2 anklets shall be vat black 94. The use of dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed and finished anklets shall contain no more labile sulfur than that shown by the standard sample when tested as specified in 4.4.4. When a standard sample is not available, the dyed and finished anklets shall show not more than a slight trace of labile sulfur when tested as specified in 4.4.4.

3.5.1 Colorfastness The dyed anklets shall show fastness to laundering, perspiration, and wet and dry crocking, equal to or better than the standard sample. When no standard sample is available, the finished anklets shall show "good" fastness to laundering and perspiration and "fair" fastness to wet and dry crocking. The Munsell value for wet and dry crocking shall not be lower than 6.5 when tested as specified in 4.4.4.

3.5.2 Matching The color shall match the standard sample under natural daylight, or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent light at 2800° Kelvin.

3.5.3 Whiteness retention The class 1 anklets shall not discolor to a degree exceeding that shown by the standard sample when tested as specified in 4.4.4.

*3.6 Pairing The anklets shall be matched and paired so that there will be no greater difference in foot length than 1/4 inch and no greater difference than 1/2 inch in leg length. Each pair of anklets shall be secured together at the tops of the leg by a pressure sensitive paper label. The label's backing substance shall not adhere to the anklets when the label is removed. The use of thread, twine, or staple to secure the pair of anklets will not be permitted.

3.7 Finishing The class 1 anklets shall be scoured, bleached, and boarded on forms. The class 2 anklets shall be scoured, dyed and boarded on forms.

*3.8 Sizes and measurements The measurements of the finished anklets shall conform to the dimensions specified in Table I. All measurements shall be made with anklets laid out smooth, flat, and under no tension as shown in figure 1. All measurements and tolerances are expressed in inches.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:59Z
Check the source to verify that this is the current version before use.

Table I - Finished measurements

Size	Measurement		Tolerance
	Foot length (A)	Leg length (B)	
8 1/2	8 1/2	8	+ 3/8 - 1/4
9	9	9	+ 3/8 - 1/4
9 1/2	9 1/2	9	+ 3/8 - 1/4
10	10	9 1/2	+ 3/8 - 1/4
10 1/2	10 1/2	10	+ 3/8 - 1/4
11	11	10 1/2	+ 3/8 - 1/4
11 1/2	11 1/2	10 1/2	+ 3/8 - 1/4

NOTE: A and B refer to Figure 1.

A. Measurement taken from the tip of the toe through the bottom of the gore of the heel to a point at the back of the heel.
B. Measurement taken from bottom edge of heel through gore, parallel to ribbed portion and extending to top edge of anklet.

3.9 Marking Each anklet shall have a size marking and fiber content marking applied just below the knit-rib instep and centered between the ankle and toe. The marking shall be in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act. Size numerals shall be not less than 5/16 inch high. All other letters and numerals shall be not less than 1/16 inch high. All markings shall be clearly legible.

3.10 Workmanship The finished anklets shall conform to the quality established by this specification. The occurrence of defects shall not exceed the limits set by the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:59Z
Check the source to verify that this is the current version before use.

4.3 First article inspection The first article submitted in accordance with 3.2 shall be inspected as specified in 4.4.2.1 and 4.4.2.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.4.1 Component and material inspection In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, a certificate of compliance shall be submitted for the requirements specified in 3.3.1, 3.3.2, 3.3.3 and 3.3.4.

4.4.2 Examination of the end item The defects found during the examination of the end item shall be classified in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The sample unit shall be one anklet. The lot size shall be expressed in units of one anklet.

4.4.2.1 Visual examination The finished anklets shall be examined for defects in design, material, construction, workmanship, and marking and classified in accordance with the defects listed below.

Classification of Defects

Defect 1/	Classification	
	Major	Minor
Rubber (covered) welt end missing	X	
Any hole, runs, or dropped stitches, including holes in looping or seaming	X	
Thin areas or mend	X	
Welt (selvage edge) omitted, cut, frayed or raveled	X	
Welt (selvage edge) uneven or not clearly defined		X
Thin yarn		X
Loose course		X
Poor looping, or seaming or fashioning of heel and toe resulting in tight tension, lumps, and dog ears which will not level out	X	
Looping or seaming end missing	X	
The free end of the looper or sewing thread trimmed less than 1/2 inch or more than 1 inch		X
Patterns of knit not as specified throughout entire anklet	X	
Ring toe omitted	X	
Non-uniform in overall appearance, not well shaped		X
Slubby yarn		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:59Z
Check the source to verify that this is the current version before use.

Defect 1/	Classification Major	Minor
Streak, stain, shade bar, or spot, clearly visible		X
Off shade or mismatched for shade		X
Rancid or bad odor	X	
Not boarded properly affecting appearance and dimension		X
Stretch lacking (tight) or loosely knit (slack or sleazy)	X	
Markings - burn or scorch in fabric	X	
Markings - missing, incorrect, illegible or incomplete		X
Reinforcement yarn missing (toe and heel)	X	

1/ For knitted fabric defect definitions, see MIL-STD-1491.

4.4.2.2 Dimensional examination The anklets shall be examined for pairing and measurement defects described below. The lot size shall be expressed in units of one (1) anklet. Any anklet deviating from measurements specified in 3.6 and 3.8 shall be scored as a defect.

4.4.2.2.1 Pairing examination A defect shall be scored if a pair of anklets is mismated for overall foot length by more than 1/4 inch or leg length by more than 1/2 inch. The sample unit for this examination shall be one pair of anklets.

4.4.2.2.2 Measurement examination Examination for measurements shall be in accordance with the defects listed below. The sample unit for this examination shall be one individual anklet. Each anklet selected for the pairing examination shall be examined for measurement defects.

	Defects
Leg length	Not within specified tolerances
Foot length	Not within specified tolerances

4.4.2.3 Inspection levels and acceptable quality levels The inspection levels and acceptable quality levels (AQL's), expressed in defects per one hundred units (DHU) shall be as follows:

	Inspection level	AQL's Major	Total
For defects applicable to 4.4.2.1	I	2.5	6.5
For defects applicable to 4.4.2.2	S-4	(one class)	4.0
For defects applicable to 4.4.3	S-2	(one class)	2.5

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:59Z
Check the source to verify that this is the current version before use.

4.4.3. Examination of packaging An examination shall be made to determine that packaging, packing and marking complies with Section 5 requirements of this specification. Defects shall be scored in accordance with the list shown below. The sample unit shall be one shipping container fully packed, selected prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the end item inspection lot.

Examine	Defect
Marking (exterior) and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Material	Any component missing or not as specified. Any component damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling of improper taping. Bulging or distortion of containers.
Content	Number of anklets per bundle or container is more or less than specified.

*4.4.4 Testing of the end item The finished anklets shall be tested for the characteristics listed in Table II. Results are applicable to the average of the determinations made on the sample unit. The sample unit for weight determinations shall be one dozen anklets from each size in the lot. The sample unit for all other tests shall be two pairs of anklets (any size). The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. Methods of testing specified herein and in FED-STD-191 shall be followed. Sampling for testing shall be in accordance with the following:

Lot size (lot units)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Testing of end item

Material Characteristic	Specification Requirement	Reference Test Method
Total no. needles	3.4	Visual
Weight per dozen (conditioned)	3.4.1	Std Comm'l
Colorfastness to:		
Laundering	3.5	5610
Perspiration	3.5	5680
Crocking 1/	3.5	5651
Discoloration (whiteness retetion) (class 1)	3.5.3	5660 2/ 3/
Presence of labile sulfur	3.5	2020

1/ Method "B" evaluation.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:59Z
Check the source to verify that this is the current version before use.

2/ One determination shall be made on each sample unit and the result reported as "pass" or "fail".
3/ Method 5660 shall be used except that exposure of the specimen and the standard sample shall be 20 standard fading hours and the exposed samples shall be allowed to lie in the dark at room temperature for a minimum of 48 hours before comparison for discoloration is made. Any specimen showing discoloration not greater than that of the standard sample shall be reported as "pass". Any specimen showing discoloration greater than that of the standard sample shall be reported as "fail". One determination shall be made for each sample.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A Each pair of anklets shall be folded at the heels in accordance with the industry's practice. Ten pairs of folded anklets of one size and class only, with each pair reversed, or every five pairs reversed, shall be evenly stacked in a bundle and securely tied at both ends with cotton tape or twine.

5.1.2 Level C (Commercial Packaging) Anklets shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets this requirement.

5.2 Packing Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A One hundred twenty pairs of anklets of one size and class only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to class weather-resistant, grade V2s, size 1A or MIL-B-17757. Level A packages shall be packed within the shipping container in the following manner: Bottom layer: (1) place four bundles against one side panel, with the length of the bundles parallel to the end panels of the container. (2) place two bundles flat in the remaining space at the opposite side panel with the length of the bundles parallel to the side panels of the container. Top layer: Repeat (1) and (2), except that the layout shall be reversed. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or order, or when there is less than the required amount per container of the same size, mixed sizes of one class only may be packed in the same shipping container.

5.2.2 Level B One hundred twenty pairs of anklets of one size and class only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, grade 275, size 1A of MIL-B-17757. Level A packages shall be packed within the shipping container in the following manner: Bottom layer: (1) place four bundles flat against one side panel, with the length of the bundles parallel to the end panels of the container. (2) Place two bundles flat in the remaining space at the opposite side panel with the length of the bundles parallel to the side panels of the container. Top layer: Repeat (1) and (2), except that the layout shall be reversed. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or order, or when there is less than the required amount per container of the same size, mixed sizes of one class only may be packed in the same shipping container.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:59Z
Check the source to verify that this is the current version before use.

5.2.2.1 Weather-resistant fiberboard containers When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with MIL-B-17757, and closed in accordance with the appendix of the box specification.

*5.2.3 Level C (Commercial Packing) Anklets, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

*5.3.1 Labels, mixed sizes Each shipping container packed with mixed sizes shall have securely attached to the end directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly printed thereon and under these words shall be legibly printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use The anklets described in this specification are intended to be used by female personnel of the Department of Defense.

6.2 Ordering data Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size and class required (see 1.2).
(c) First article requirements (see 3.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

*6.3 Samples For access to samples, address the procuring activity issuing the invitation for bids.

*6.4 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the maringal notations and relationship to the last previous issue.

Custodian:
Navy - NU
Army - GL
Air Force - 11

Preparing activity:
Navy - NU

Project No. 8445-0067

Review activities:
Army - MD
Air Force - 99
DLA - CL

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:59Z
Check the source to verify that this is the current version before use.

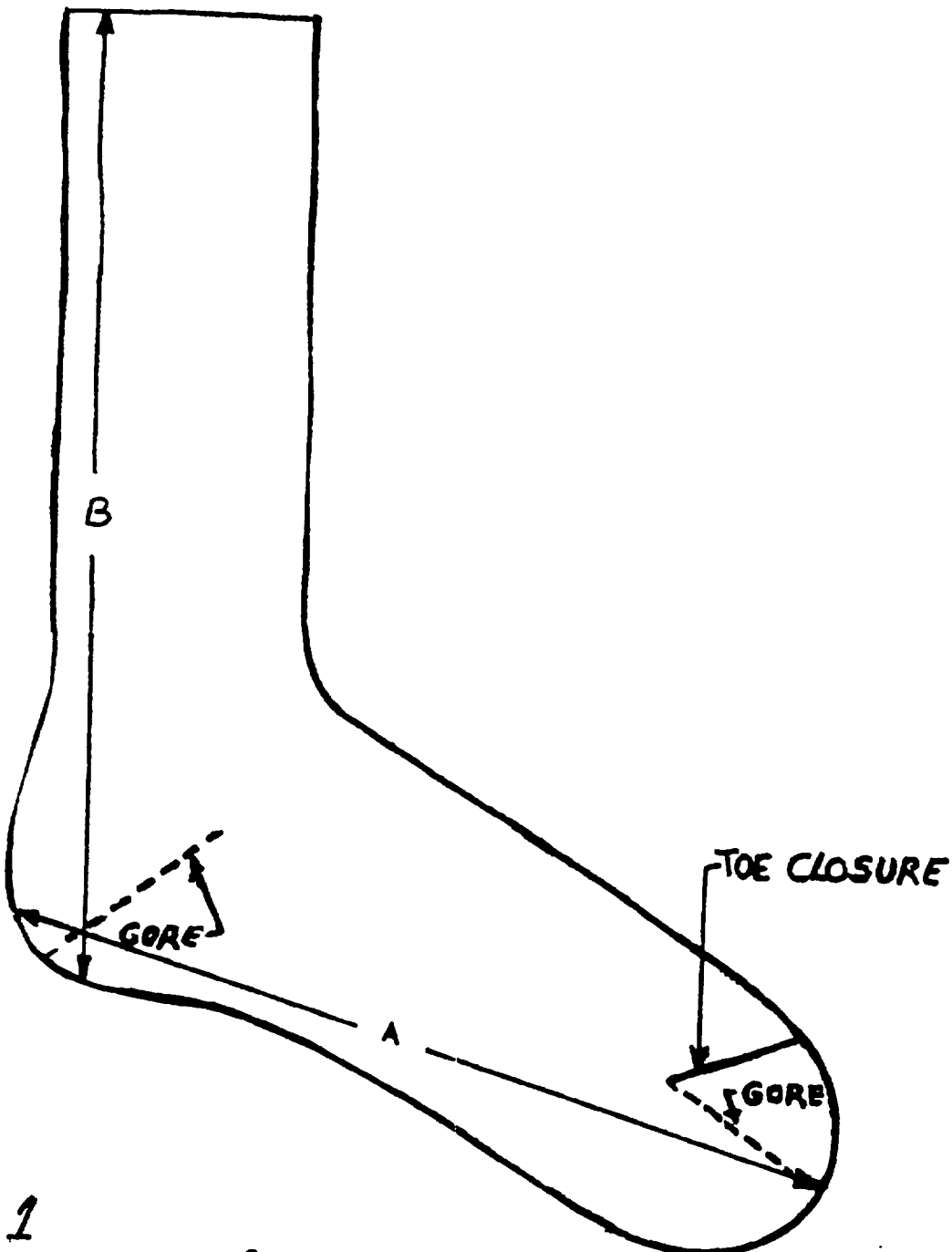


FIGURE 1
ANKLETS, WOMEN'S: COTTEN

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:59Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:59Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-A-22789C ANKLETS, WOMEN'S: COTTON

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 202640-19T13:59Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Officer In Charge
Navy Clothing and Textile Research Facility
21 Strathmore Road
Natick, MA 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:59Z
Check the source to verify that this is the current version before use.